

FCA(FUNDAMENTALS OF COMPUTER APPLICATIONS

MODULE-1

Author: Algae Services

Decade of experience in Corporate Training Classroom sessions in different streams of Technology.

Expertise in Database Technologies, Analytics, Big Data, ERP, Business Process Re-Engineering, Data modeling.

Worked with top service and product companies like Wipro , Volvo , TVS , TEG Analytics, General Electrics , Royal Bank of Scotland , Verizon etc.

Worked with Universities like Praxis, IIDT, Snt. Claret, Matrushi, Jain University , RGTU

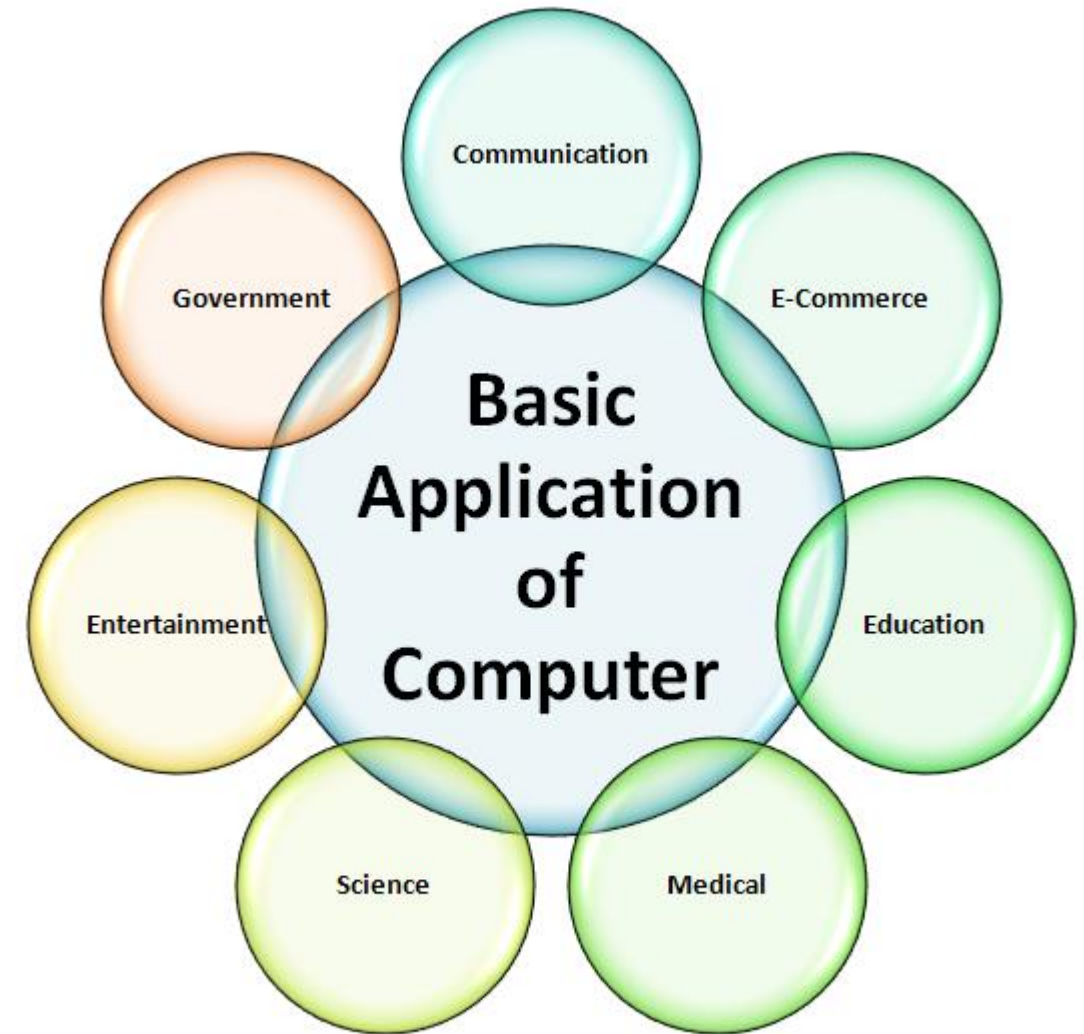
Basics of Computer System:

- Concept of Hardware and Software,
- Computer block diagram, Input Output unit, CPU, Control Unit, Arithmetic logic Unit (ALU), Memory Unit, Monitor.
- Printers: Dot matrix, Laser, Inkjet, Plotters, Scanner.
- System software and Application Software,
- Operating system concepts, purpose and functions, Operations of Windows OS.
- Creating and naming of file and folders,
- copying file, renaming and deleting of files and folders, searching files and folders, installation application, creating shortcut of application on the desktop
- Overview of control Panel, Taskbar.

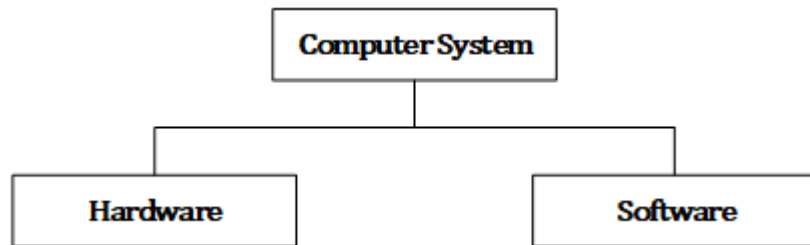
Basics of Computer System:

INTRODUCTION

- A computer system primarily comprises a central processing unit (CPU), memory, input/output devices and storage devices. All these components function together as a single unit to deliver the desired output. A computer system comes in various forms and sizes
- A computer is an electronic device, operating under the control of instructions stored in its own memory that can accept data (input), process the data according to specified rules, produce information (output), and store the information for future use¹. Functionalities of a computer.
- An application, also referred to as an application program or application software, is a computer software package that performs a specific function directly for an end user or, in some cases, for another application. An application can be self-contained or a group of programs.



Computers Have Two Main Parts



Computer Hardware
Computer Software



HARDWARE

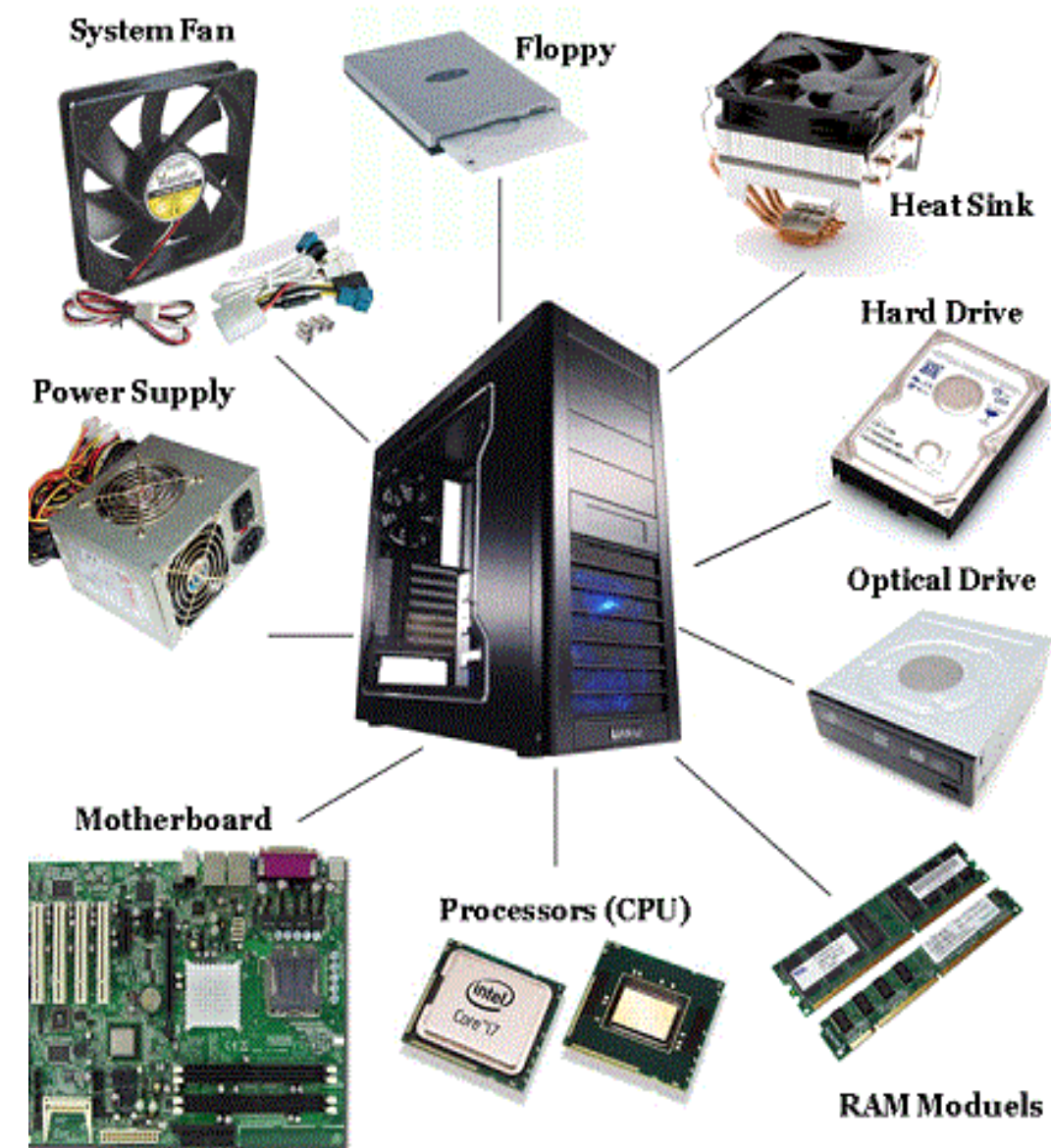


SOFTWARE

- computer hardware is the physical components that a computer system requires to function.
- It encompasses everything with a circuit board that operates within a PC or laptop ,CPU, mother board. Floppy disc...etc

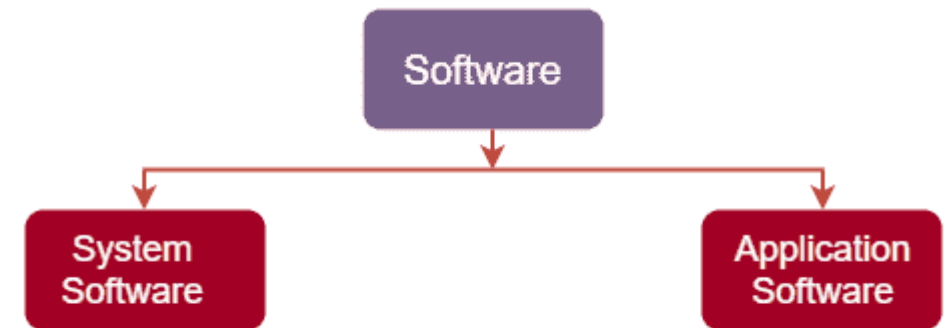
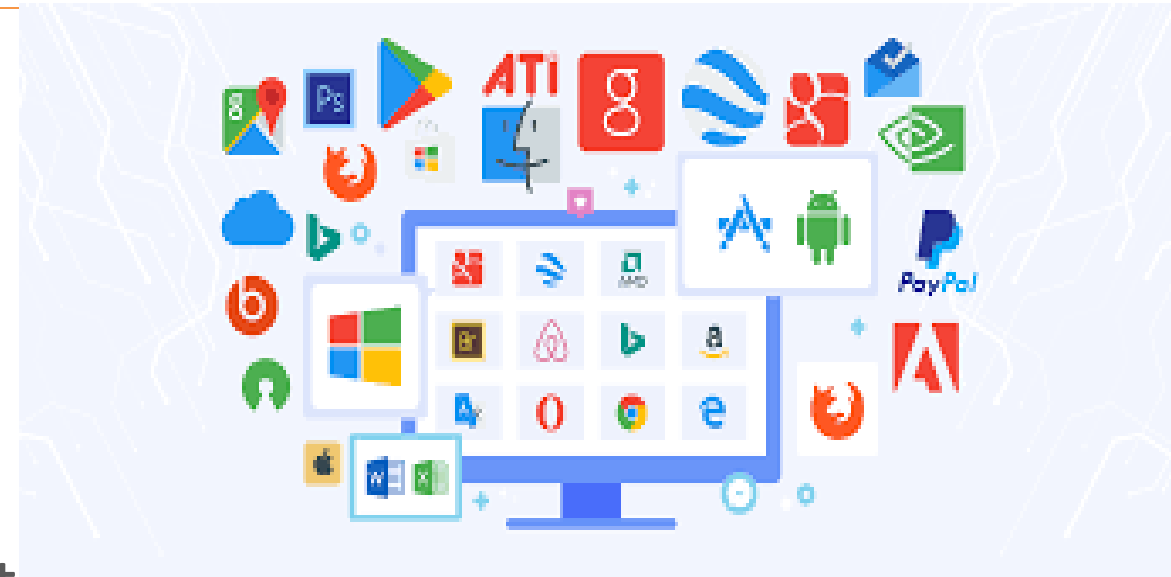
There are 5 main hardware components

1. **Input devices** -keyboard, mouse, etc.
2. **Processing-** CPU,GPU, Mother board, etc
3. **Storage devices** - RAM,ROM,FLOPPY DISC,etc
4. **Output devices** -printer, monitor, etc.
5. **Communication devices** – Smart Phone, Bluetooth, wi-fi , Routers



- Software is a set of instructions, data or programs used to operate computers and execute specific tasks.
- Computer Software programs are instructions for your computer.
- These instructions allow you to do different types of tasks.
- Software is sometimes called an application or a program.
- example :- Google Chrome.

Mozilla Firefox, Internet Explore.,
Opera, UC Browser, Safari.



Types of Software

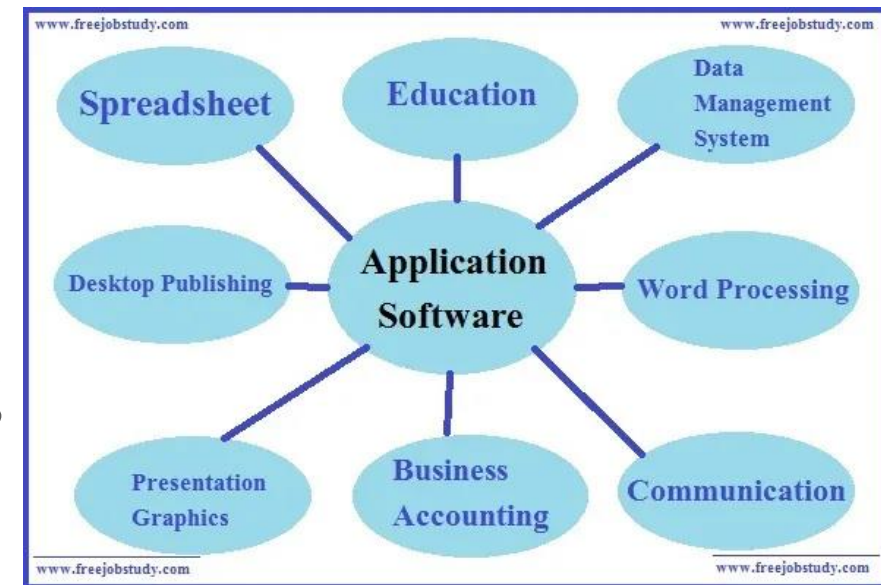
- System software is a type of computer program that is designed to run a computer's hardware and application programs.

Ex- mac os, Googlechrome, skype,...etc

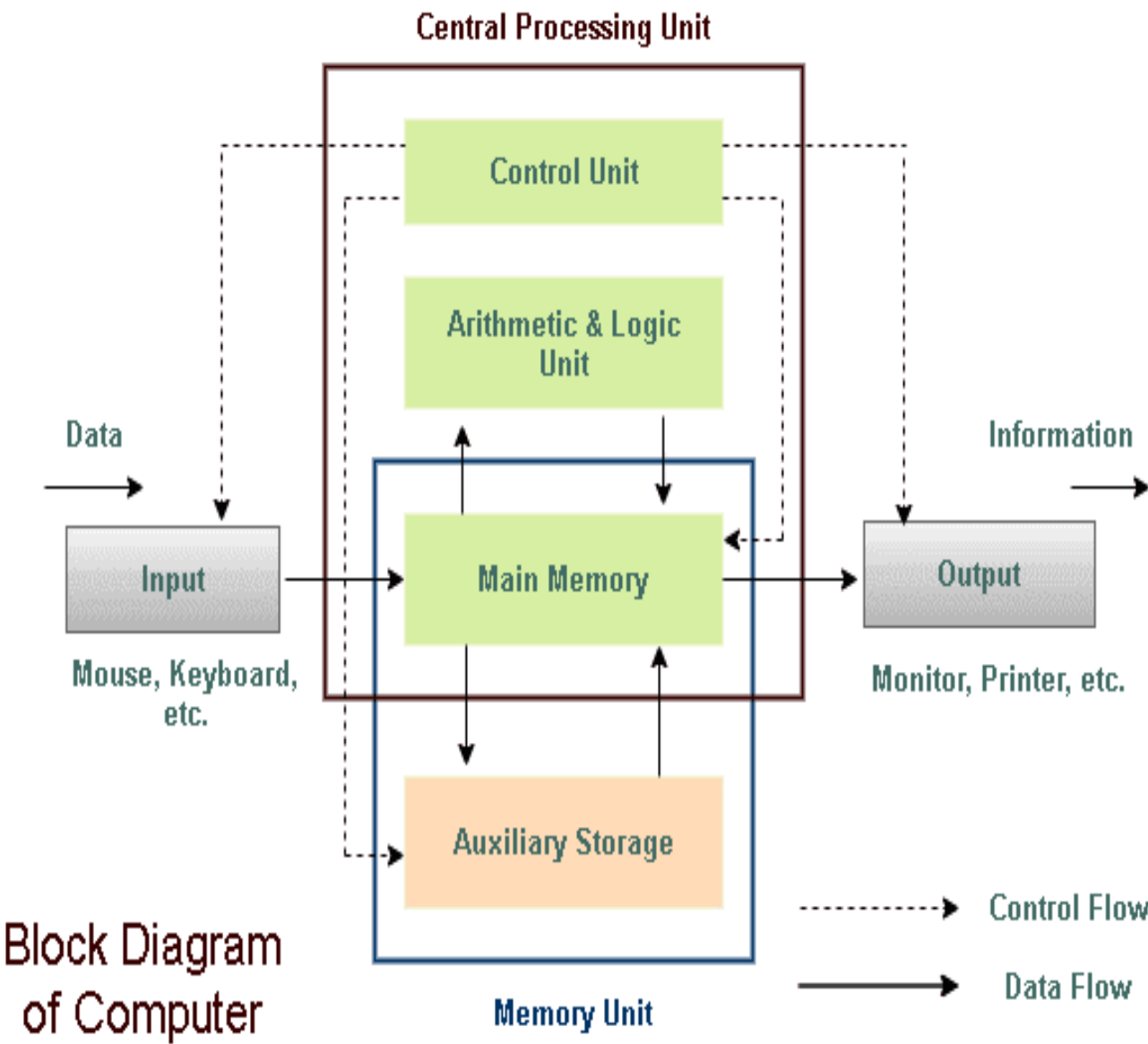
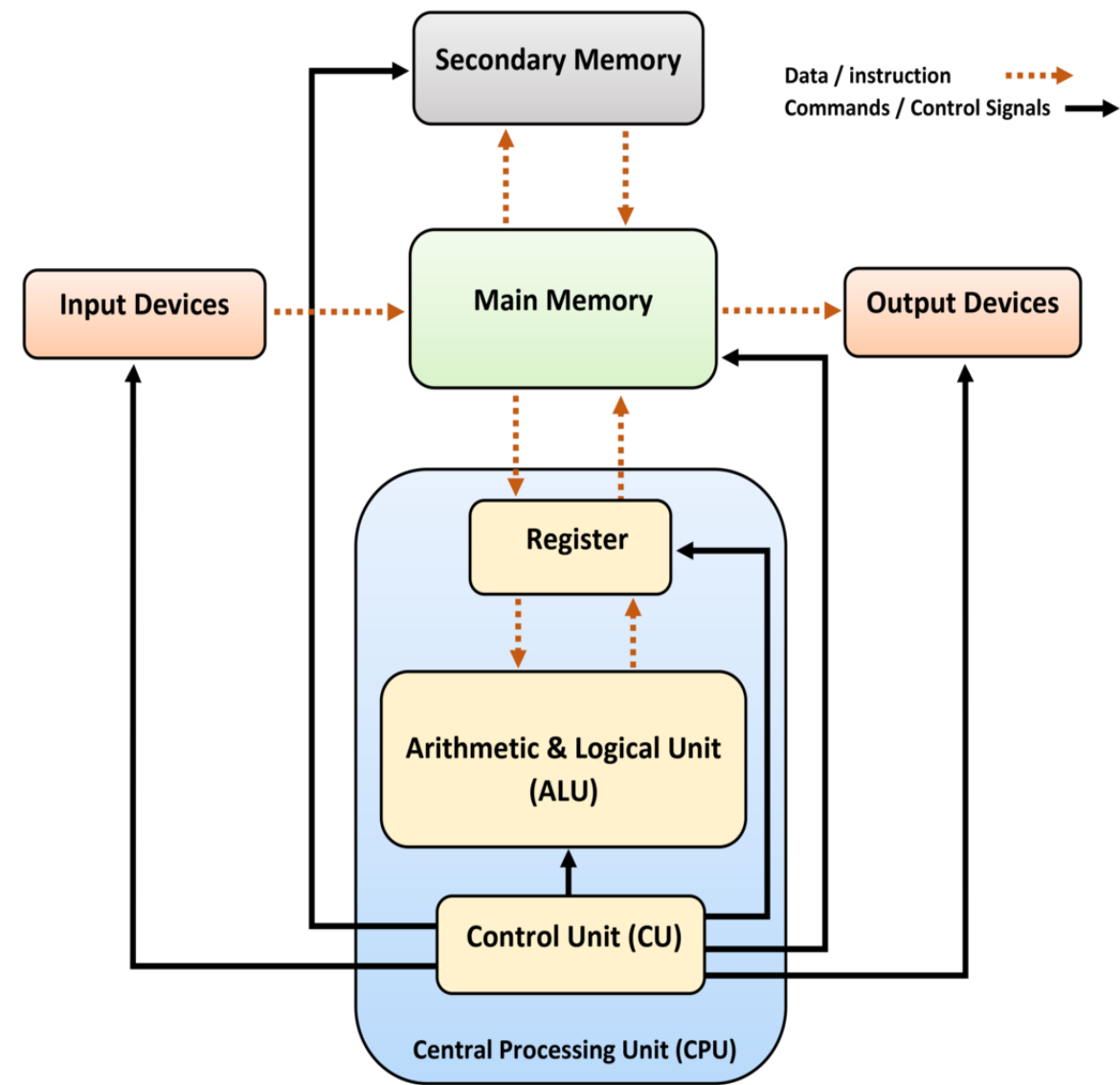
- Application software is a type of computer program that performs a specific personal, educational, and business function.

Ex;-Microsoft products such as Office, PowerPoint, Word, Excel, Outlook, etc.

Application software enables the user to complete tasks such as creating documents, spreadsheets, databases, and publications, doing online research, sending email, designing graphics, running businesses, and even playing games.



Hardware	Software
It is the physical component of a computer system.	It is the programming language that makes hardware functional.
It has the permanent shape and structure, which cannot be modified.	It can be modified and reused, as it has no permanent shape and structure.
The external agents such as dust, mouse, insects, humidity, heat, etc. can affect the hardware (as it is tangible).	The external agents such as dust, mouse, insects, humidity, heat, etc. cannot affect (as it is not tangible).
It works with binary code (i.e., 1's to 0's) .	It functions with the help of high level language like COBOL, BASIC, JAVA, etc.
It takes in only machine language, i.e., lower level language.	It takes in higher level language, easily readable by a human being.
It is not affected by the computer bug or virus.	It is affected by the computer bug or virus.
It cannot be transferred from one place to other electronically.	It can transfer from one place to other electronically.
Duplicate copy of hardware cannot be created.	A user can create copies of a software as many as he wishes.



1. Computers need to receive data and instruction in order to solve any problem. Therefore, we need to input the data and instructions into the computers.
2. The input unit consists of one or more input devices. Keyboard is the one of the most commonly used input device.
3. an input device, a piece of computer hardware apparatus used to supply a data processing system including a computer or information device with control and data signals.
 - Accept the data and instructions from the outside world.
 - Convert it to a form that the computer can understand.
 - Supply the converted data to the computer system for further processing.



1) Keyboard

The [keyboard](#) is a basic input device that is used to enter data into a computer or any other electronic device by pressing keys. It has different sets of keys for letters, numbers, characters, and functions. Keyboards are connected to a computer through [USB](#) or a Bluetooth device for wireless communication.

2) Mouse

The mouse is a hand-held input device which is used to move cursor or pointer across the screen. It is designed to be used on a flat surface and generally has left and right button and a scroll wheel between them. Laptop computers come with a touchpad that works as a mouse.

3) Scanner

The scanner uses the pictures and pages of text as input. It scans the picture or a document. The scanned picture or document then converted into a digital format or file and is displayed on the screen as an output.



- a unit which delivers information from the computer to an external device or from internal storage to external storage.
- The output unit provides the results of computer process to the users i.e it links the computer with the external environment.
- Most of the output data is the form of audio or video. The different output devices are monitors, printers, speakers, headphones etc.
- Example-monitors and projectors (video), headphones and speakers (audio), or printers and plotters (physical reproduction in the form of text or graphics).

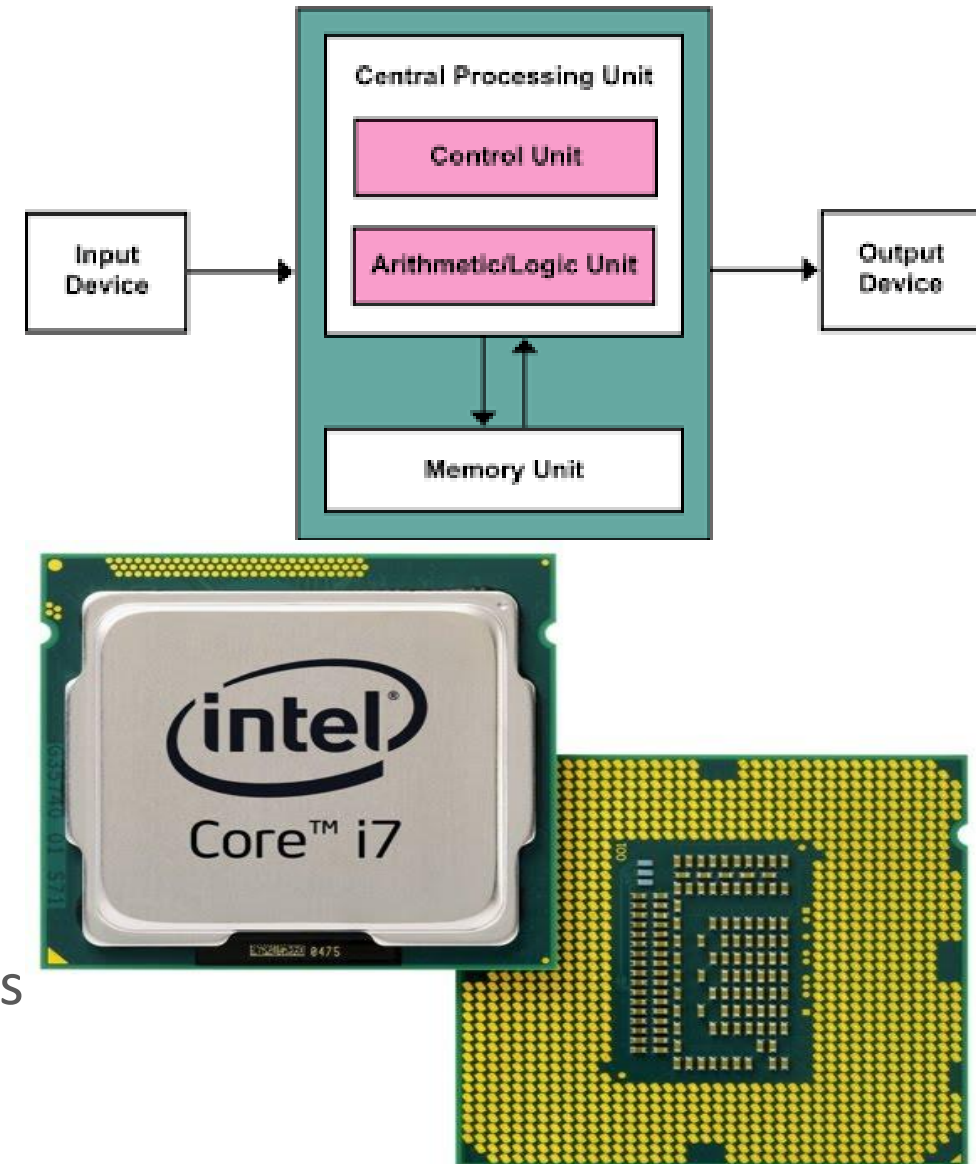


- A printer is a **computer output device that accepts text and graphic output from a computer and transfers the information to paper**
- A computer monitor is an **output device that displays information in pictorial or text form**. A monitor usually comprises a visual display, some circuitry, a casing, and a power supply.
- A computer speaker is an **output hardware device that connects to a computer to generate sound**. The signal used to produce the sound that comes from a computer speaker is created by the computer's sound card.

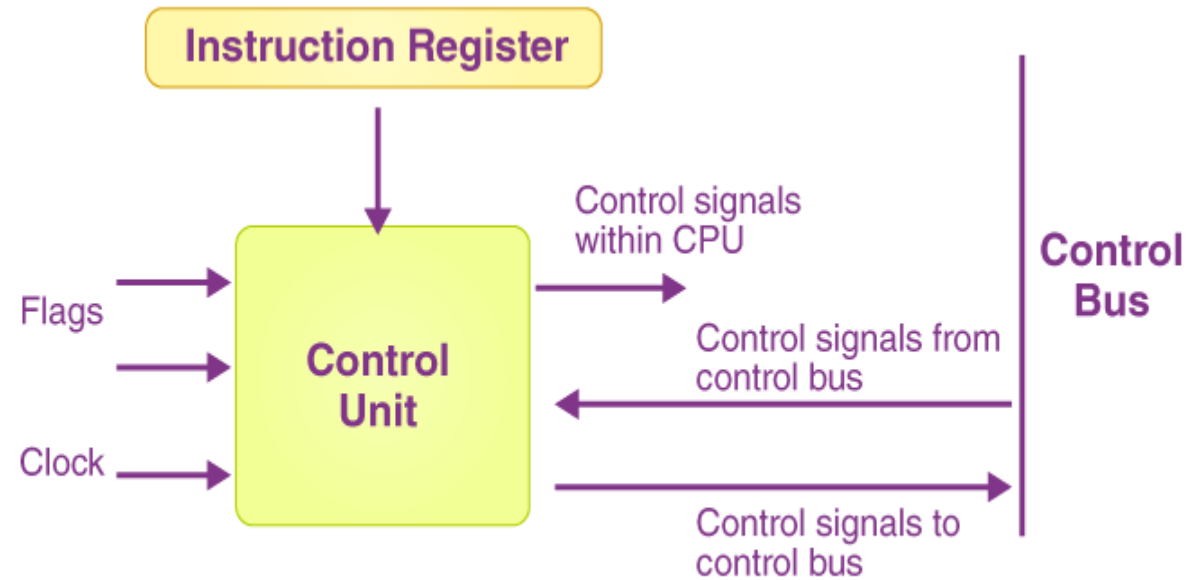


- The computer's central processing unit (CPU) is **the portion of a computer that retrieves and executes instructions.**
- The CPU is essentially the brain of a CAD system.
- The CPU is often simply referred to as the processor.
- The CPU consists of 3 major units which are:
 - a) Memory or Storage Unit,
 - b) Control Unit,
 - c) ALU (Arithmetic Logic Unit).

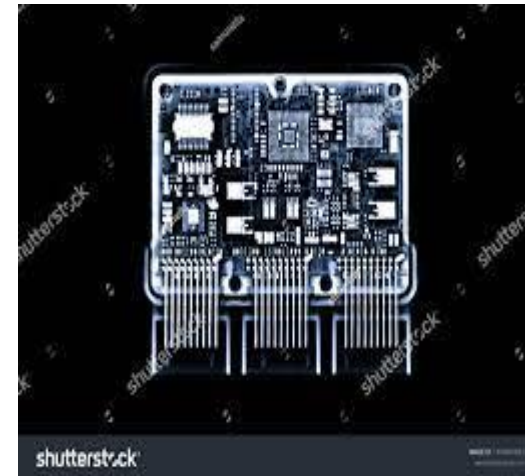
It is also known as **internal storage unit** or the **main memory** or the **primary storage** or **Random Access Memory (RAM)**. It stores all the data and the instructions required for processing. It stores intermediate results of processing.



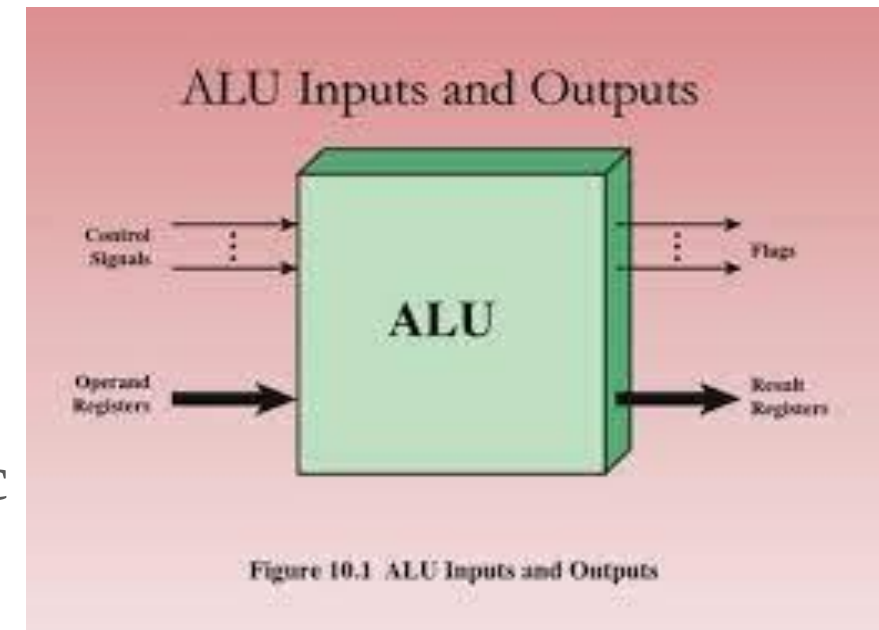
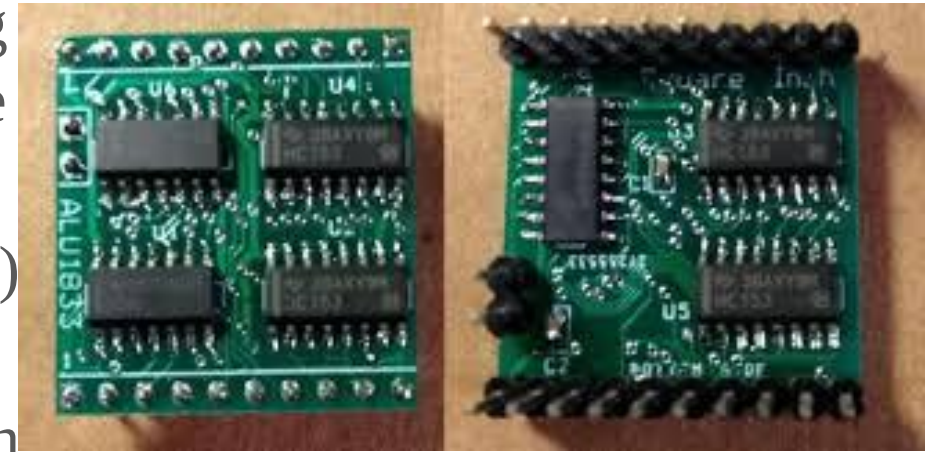
- A control unit (CU) **handles all processor control signals**. It directs all input and output flow, fetches code for instructions from micro programs and directs other units and models by providing control and timing signals.
- The control unit of the central processing unit **regulates and integrates the operations of the computer**.
- It selects and retrieves instructions from the main memory in proper sequence and interprets them so as to activate the other functional elements of the system at the appropriate moment.
- Example- CPU



Block Diagram of the Control Unit

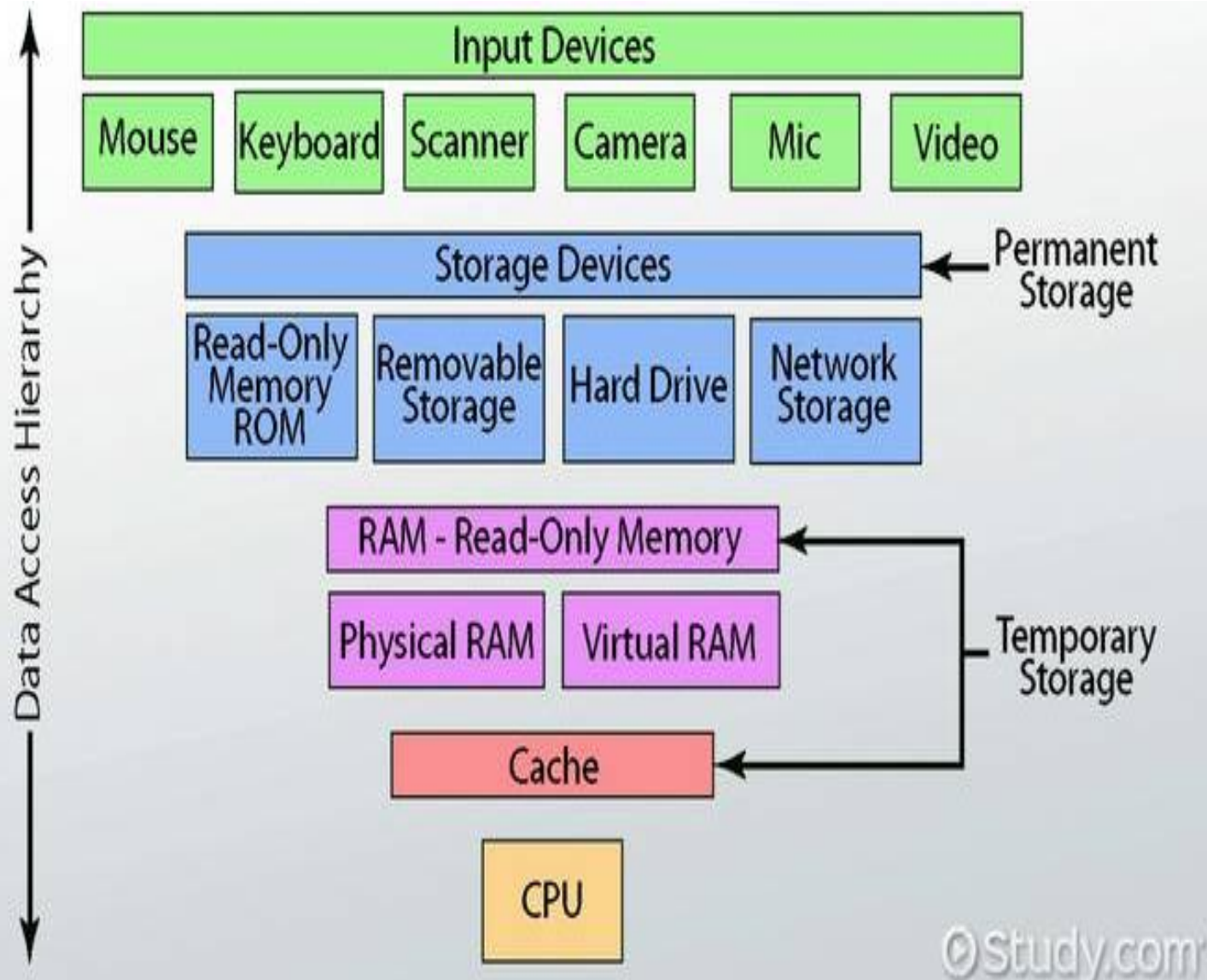


- An arithmetic-logic unit is the part of a central processing unit that carries out arithmetic and logic operations on the operands in computer instruction words.
- the ALU is divided into two units: an arithmetic unit (AU) and a logic unit (LU).
- An arithmetic unit, or ALU, enables computers to perform mathematical operations on binary numbers. They can be found at the heart of every digital computer and are one of the most important parts of a CPU (Central Processing Unit).
- Example- An ALU performs basic arithmetic and logic operations. Examples of arithmetic operations are addition, subtraction, multiplication, and division. Examples of logic operations are comparisons of values such as NOT, AND, and OR.



- Memory unit is **the amount of data that can be stored in the storage unit**. This storage capacity is expressed in terms of Bytes.
- **This unit can store instructions, data, and intermediate results**. This unit supplies information to other units of the computer when needed. It is also known as internal storage unit or the main memory or the primary storage or Random Access Memory (RAM).
- There are technically two types of computer memory: **primary and secondary**.
- Example- Read-only memory (ROM), RAM, HARD DISK, TAPE, CHIP, FLOPPY DISC..ETC





UNITS OF COMPUTER MEMORY MEASUREMENTS

1024 Bytes = 1 KB (Kilo Byte)

1024 KB = 1 MB (Mega Byte)

1024 MB = 1 GB (Giga Byte)

1024 GB = 1TB (Terra Byte)

1024 TB = 1 PB (Peta Byte)

1024 PB = 1 EB (Exa Byte)

1024 EB = 1 ZB (Zetta Byte)

1024 ZB = 1 YB (Yotta Byte)

1024 YB = 1 (Bronto Byte)

1024 Brontobyte = 1 (Geop Byte)

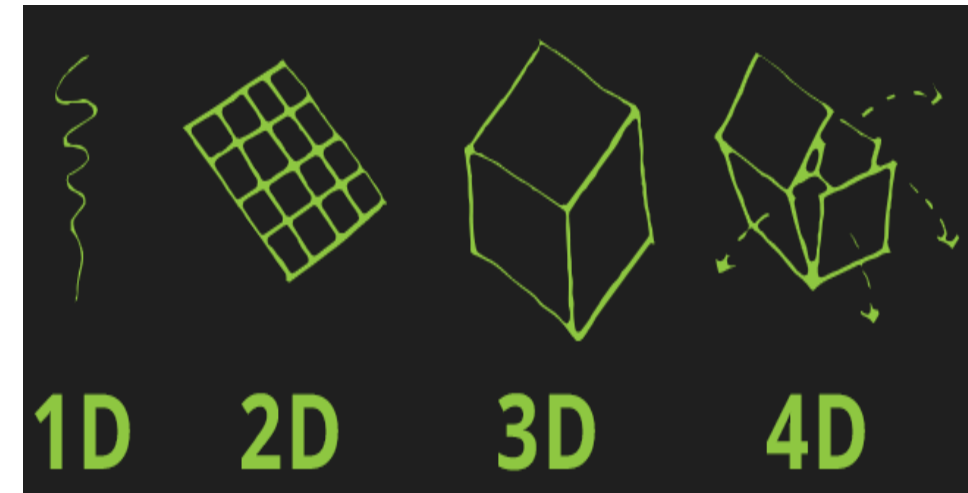
Geop Byte is the Highest Memory

- A computer monitor is **an output device that displays information in pictorial or text form**. A monitor usually comprises a visual display, some circuitry, a casing, and a power supply.
- a monitor is an output device that **displays video images and text**.
- Monitor is used for Gaming.
- Monitor is used by Business users for using various business centric software applications such as MS Excel.
- Computer Monitor is used to display the result of any data and for data entry also.
- It is use for web browsing.
- There are five types of monitors **CRT(Cathode Ray tube), LCD (Liquid Crystal Display), LED (Liquid Emitting Diode), OLED (Organic Light Emitting Diode), and Plasma Monitor** all are used in televisions or computer desktop.

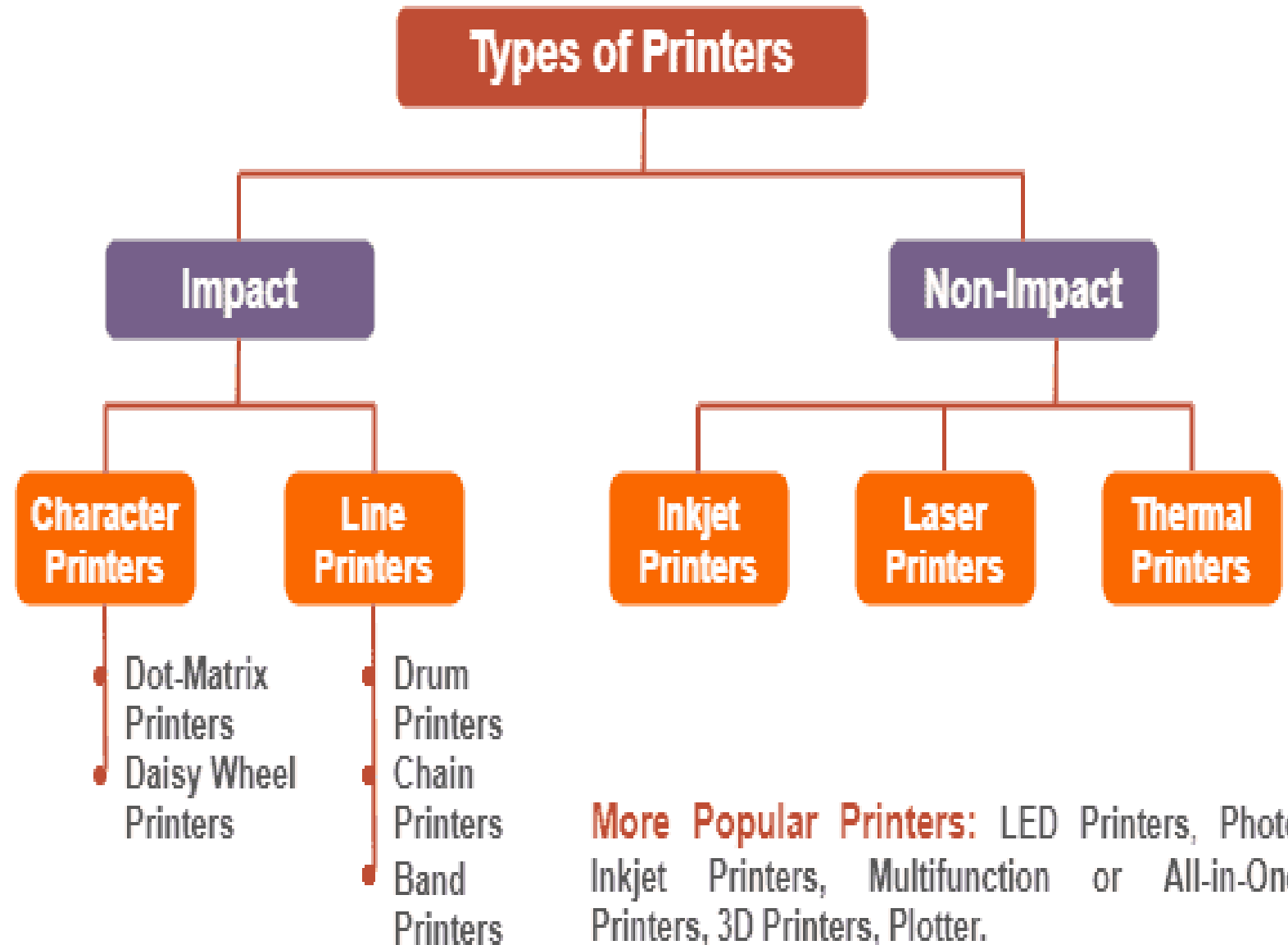


CRT monitor → LCD monitor → LED monitor → UHD monitor

- A printer is an external hardware output device that takes the electronic data stored on a computer or other device and generates a hard copy.
- A printer is a device that accepts text and graphic output from a computer and transfers the information to paper, usually to standard-size, 8.5" by 11" sheets of paper.
- Printers are one of the common computer peripheral devices that can be classified into two categories that are 2D and 3D printers.
- The 2D printers are used to print text and graphics on a paper,
- and 3D printers are used to create three dimensional physical objects.



- An impact printer has mechanisms resembling those of a typewriter.
- Example of Impact Printers, Dot-matrix printers, Daisy-wheel printers, and line printers.
- Non-Impact Printers: It is a type of printer that does not hit or impact a ribbon print.



- A dot matrix printer is an impact printer that prints using a fixed number of pins or wires.
- Dot matrix printing uses a print head that moves back-and-forth, or in an up-and-down motion, on the page and prints by impact, striking an ink-soaked cloth ribbon against the paper, much like the print mechanism on a typewriter.
- Dot matrix and inkjet printers share one key characteristic -- both make images out of small dots. With a dot matrix printer, a pin presses through a ribbon to make an impact on the page. Inkjet printers have an electrical signal that causes a microscopic quantity of ink to squirt onto the page.



- A laser printer is a popular type of computer printer that uses a non-impact photocopier technology where there are no keys striking the paper.
- Laser printers provide quality printing with sharp images and consistent black tones. They have become widely used for **printing of reports and letters and for desktop publishing**.
- Not only are laser printers faster, but they also have **higher monthly print volume**.
- Print volume is the recommended number of pages your printer can print a month for optimal performance. Monthly print volumes for laser printers are well into the thousands for most models.



HP LaserJet Printer



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- An inkjet printer is a computer peripheral that produces hard copies of a text document or photo by spraying droplets of ink onto paper.
- Inkjet Printers use a special type of ink for printing on the paper.
- Inkjet Printers are particularly used to get high quality color prints.
- They are also capable of giving instant print outputs. These machines are preferred by those people who need prints once in a while and not on a daily basis.
- Inkjet printing is a type of **computer printing that recreates a digital image by propelling droplets of ink onto paper and plastic substrates.**
- Inkjet printers were the most commonly used type of printer in 2008, and range from small inexpensive consumer models to expensive professional machines.



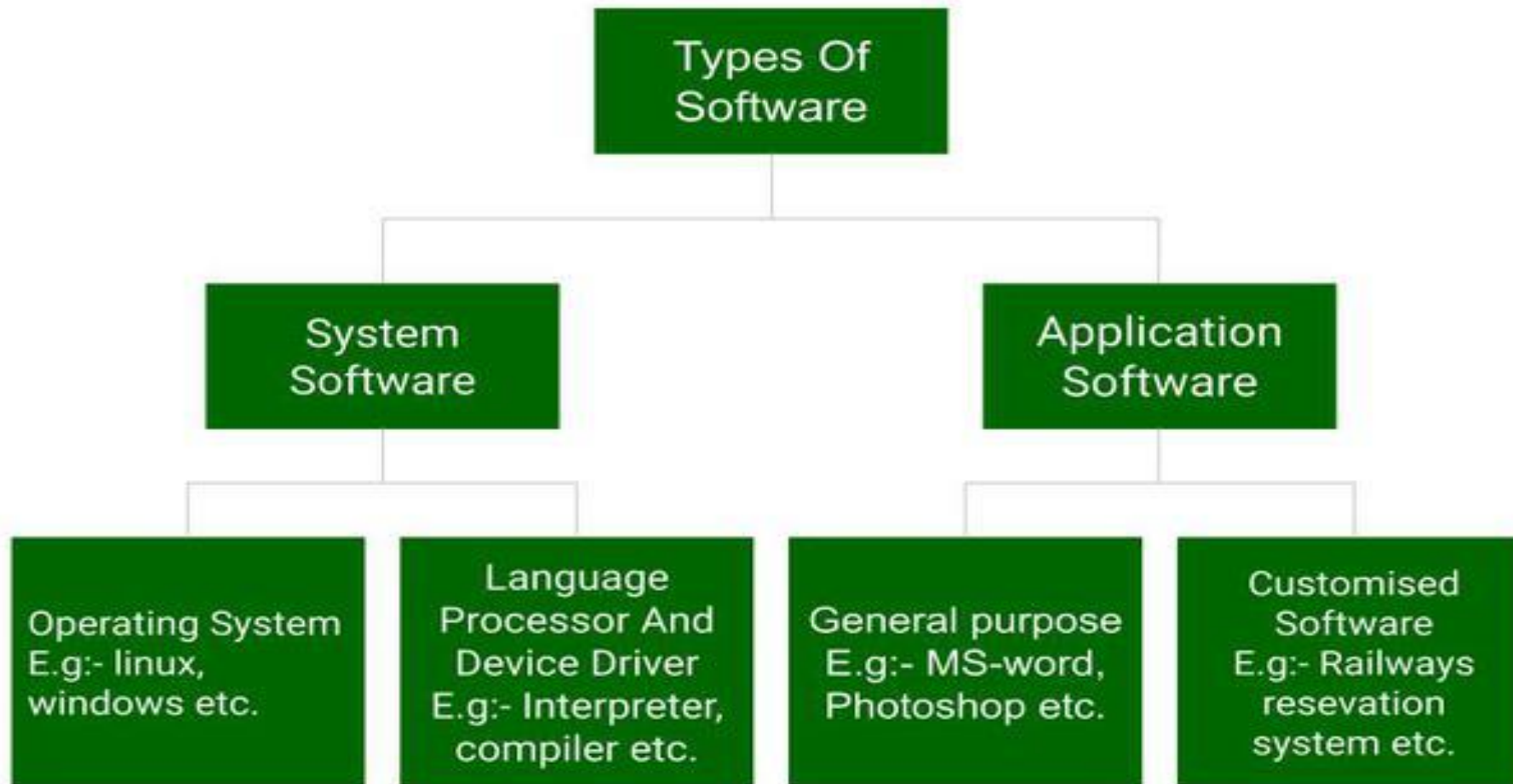
- A plotter is a printer that interprets commands from a computer to make line drawings on paper with one or more automated pens.
- Plotters are used for **all kinds of construction and MCAD applications, including technical drawings, maps, ortho photos, and renders.**
- Any application that benefits from crisp text and precise lines on a larger, single-sheet output can see value in a large format HP plotter printer.

Plotter



- A scanner is a device that captures images from photographic prints, posters, magazine pages and similar sources for computer editing and display.
- Scanners work by converting the image on the document into digital information that can be stored on a computer through optical character recognition (OCR).
- When referring to hardware, a scanner, image scanner, or optical scanner is a hardware input device that optically "reads" an image and converts it to a digital signal.
- For example, a scanner may be used to convert a printed picture, drawing, or document (hard copy) to a digital file that can be edited on a computer.





- System software is a type of computer program that is designed to run a computer's hardware and application programs.
- System software is of **three main types** :
 1. Operating system.
 2. Language processor.
 3. Utility software.

the system software is the interface between the hardware and user applications. The operating system is the best-known example of system software.



Types of software



- An operating system (OS) is the program that, after being initially loaded into the computer by a boot program, manages all of the other application programs in a computer.
- **Windows, Linux, and Android** are examples of operating systems that enable the user to use programs like MS Office, Notepad, and games on the computer or mobile phone.
- It is necessary to have at least one operating system installed in the computer in order to run basic programs like browsers.
- An operating system is the most important software that runs on a computer. It **manages the computer's memory and processes, as well as all of its software and hardware.**
- It also allows you to communicate with the computer without knowing how to speak the computer's language.

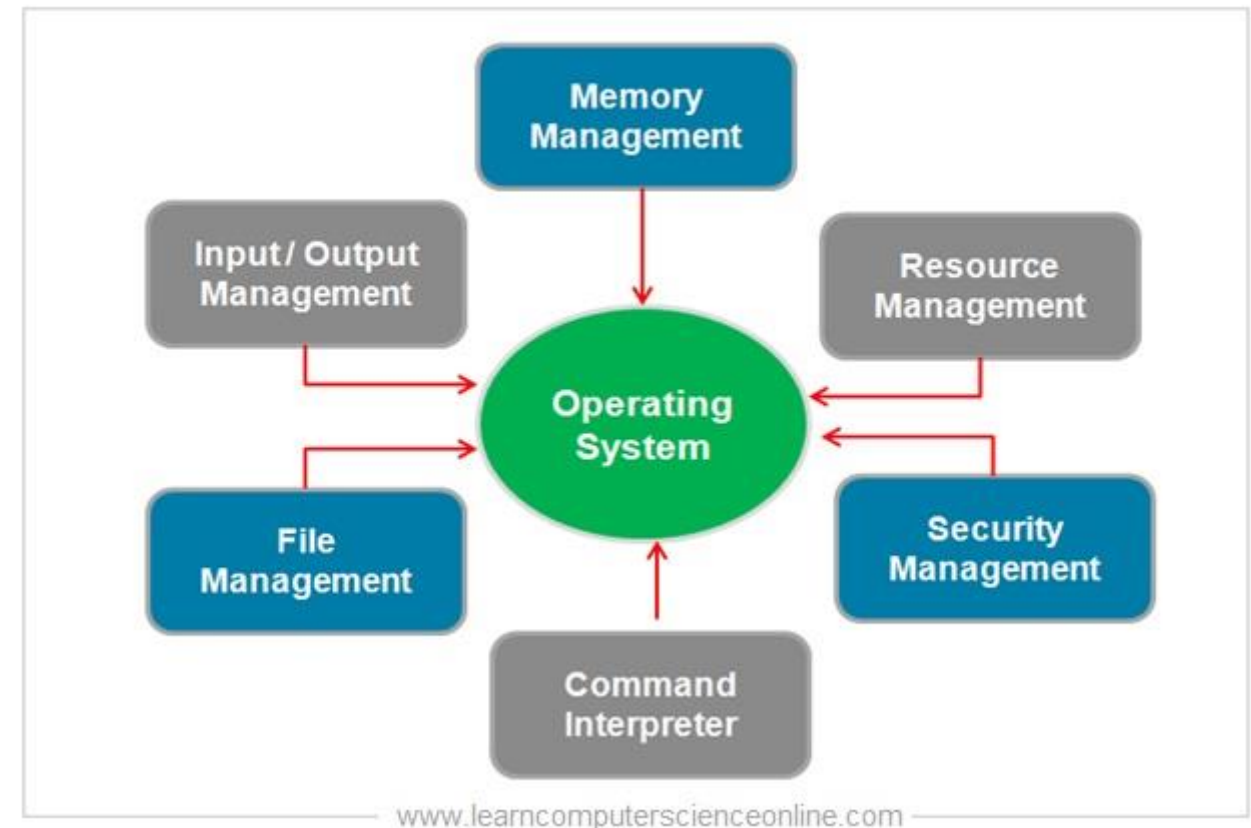


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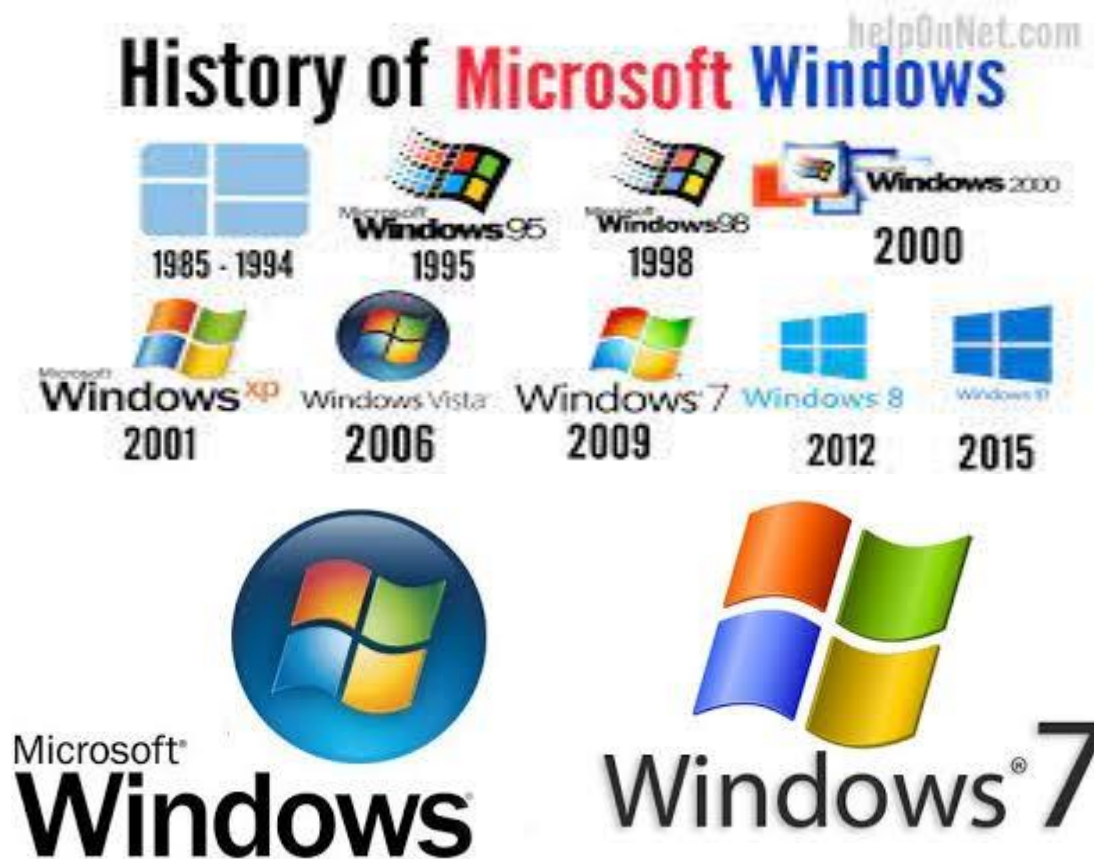
- **Functions of Operating Systems**

- a) Security. ...
- b) Control over system performance. ...
- c) Job Accounting.
- d) Error detecting aids.
- e) Coordination between other software .
- f) Memory Management. ...
- g) Processor Management. ...
- h) Device Management.

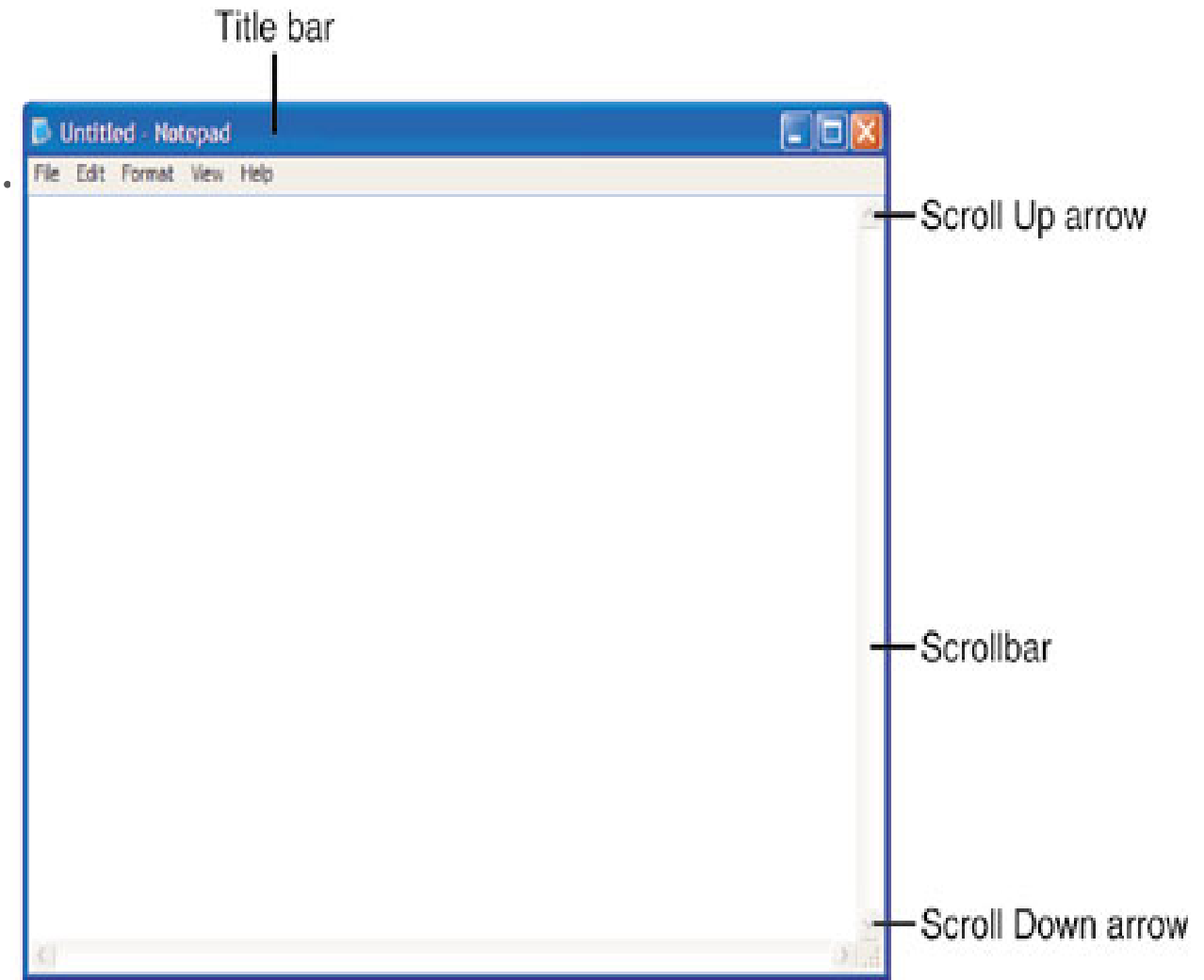
Operating System - Functions



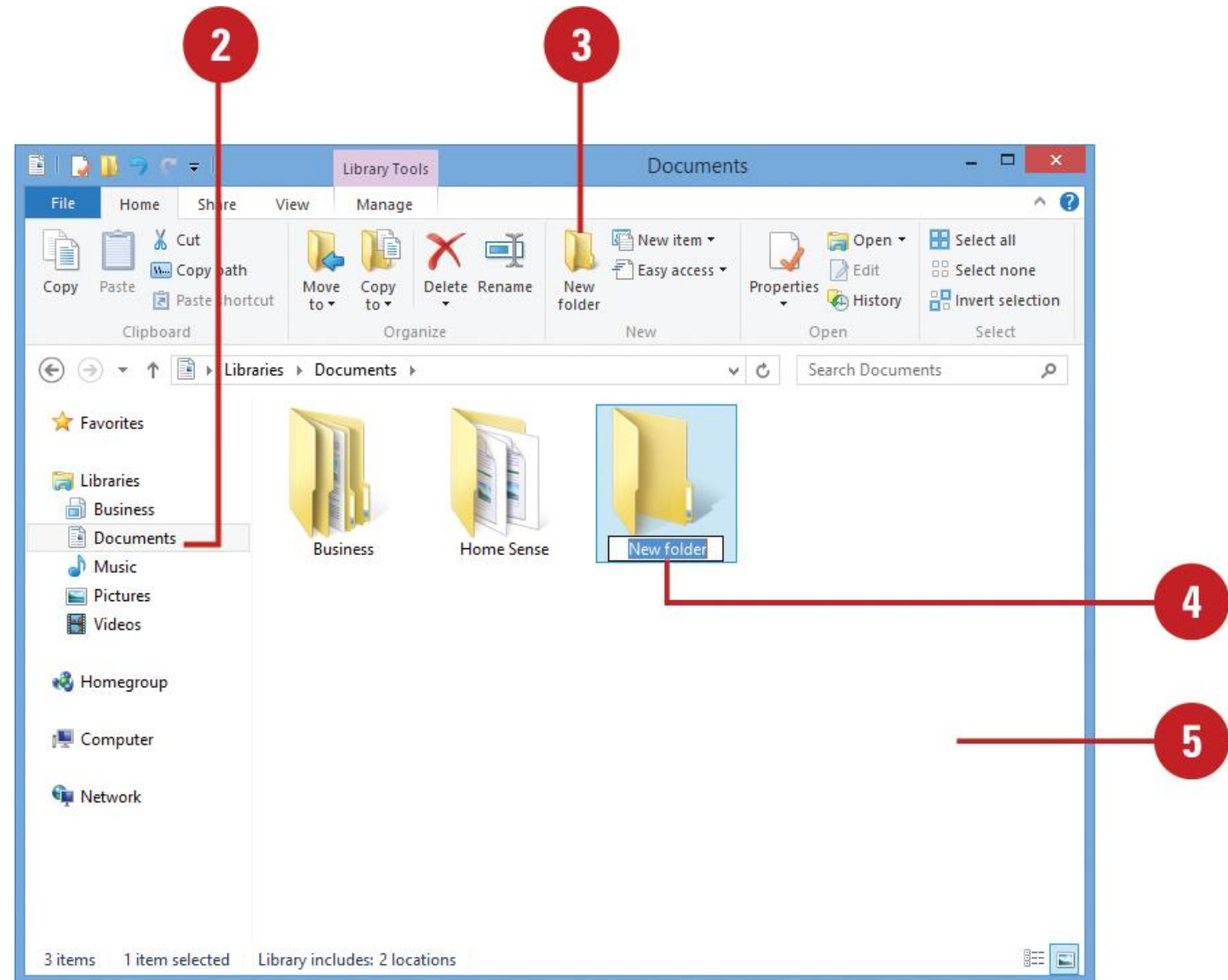
- Windows is a group of several proprietary graphical operating system families developed and marketed by Microsoft. Each family caters to a certain sector of the computing industry.
- Microsoft Windows, also called Windows and Windows OS, computer operating system (OS) developed by Microsoft Corporation to run personal computers (PCs).
- Featuring the first graphical user interface (GUI) for IBM-compatible PCs, the Windows OS soon dominated the PC market.
- **Windows 11** is the latest release of Microsoft's flagship operating system and the successor to Windows



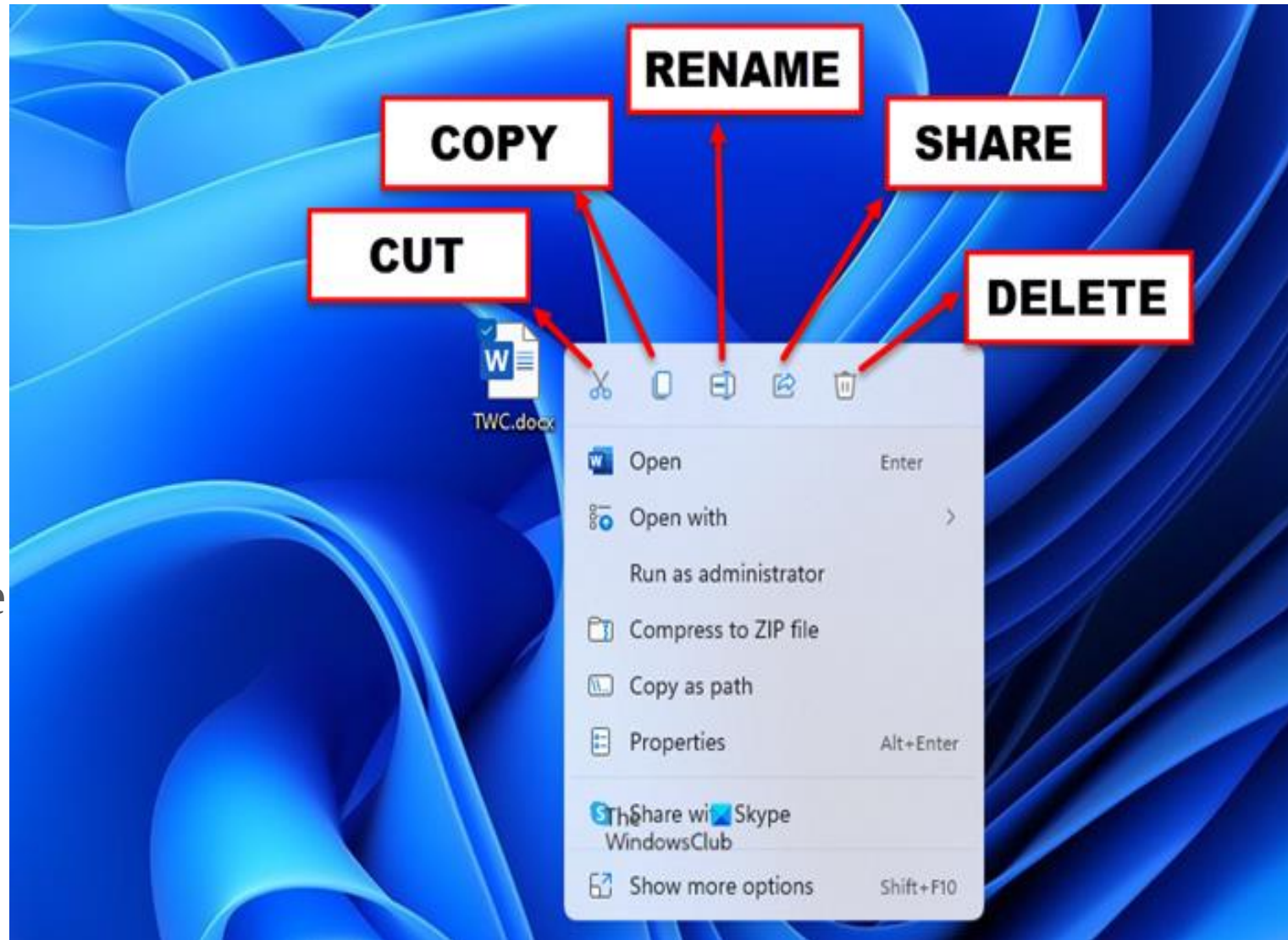
- Important Windows Operations
 - a) Pointing and Clicking. The most common mouse operation is pointing and clicking. .
 - b) Double-Clicking. ...
 - c) Right-Clicking. ...
 - d) Dragging and Dropping. ...
 - e) Hovering. ...
 - f) Moving and Resizing Windows. ...
 - g) Maximizing, Minimizing, and Closing Windows. ...
 - h) Scrolling Through a Window.



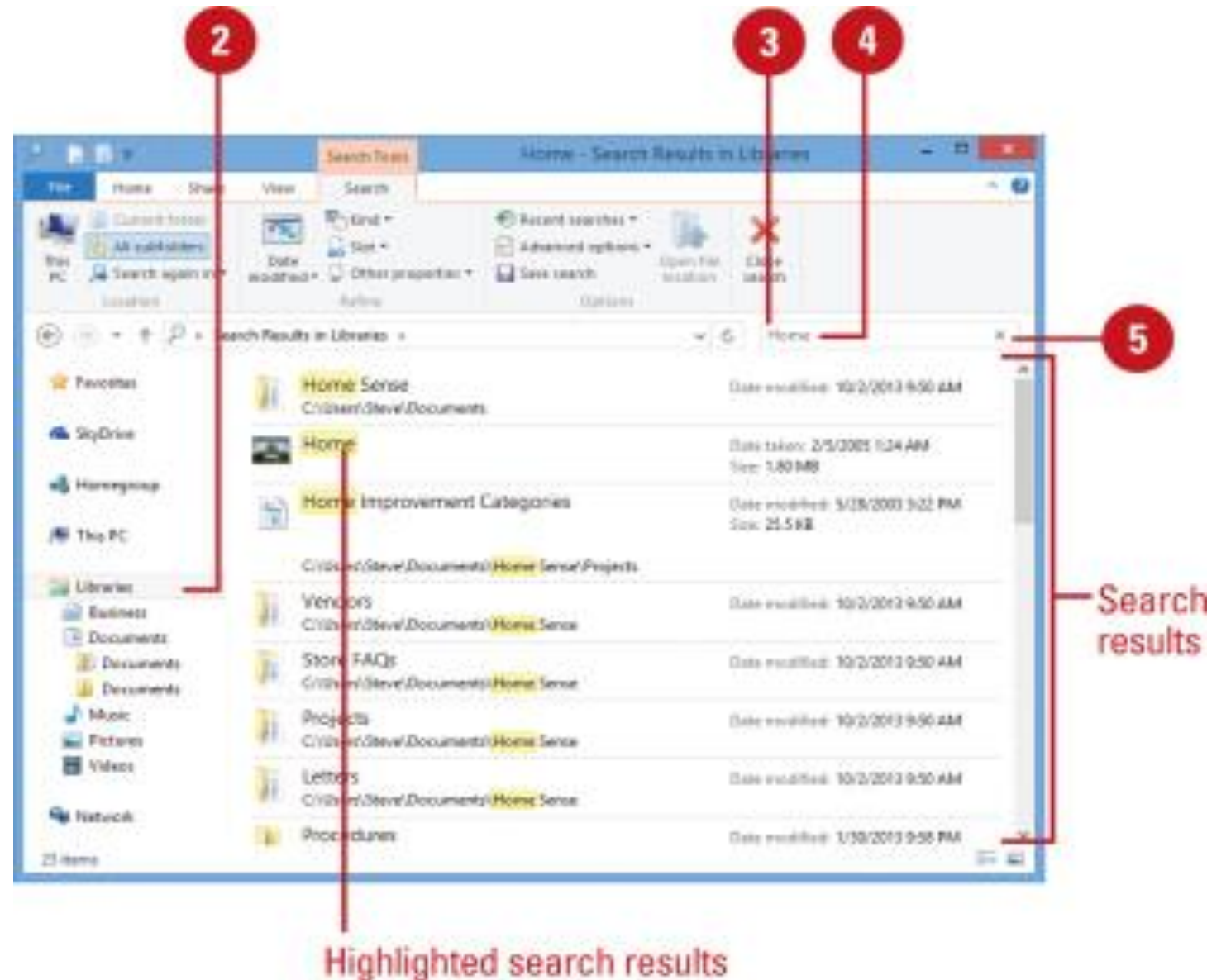
- With your document open, click File > Save As.
- Under Save As, select where you want to create your new folder. ...
- In the Save As dialog box that opens, click New Folder.
- Type the name of your new folder, and press Enter. ...
- Click Save.



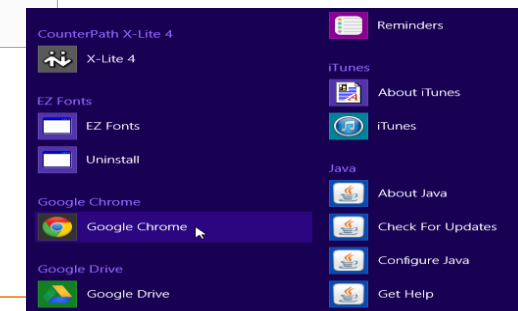
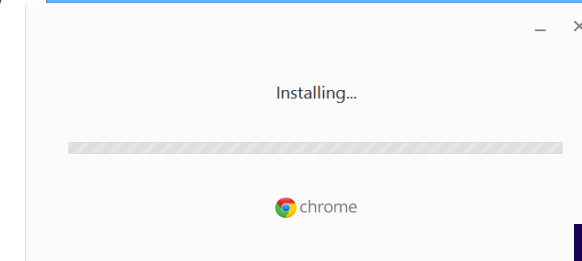
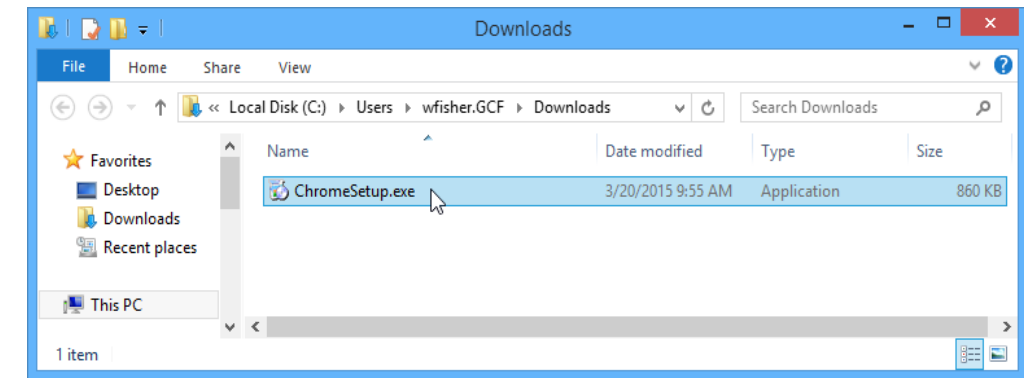
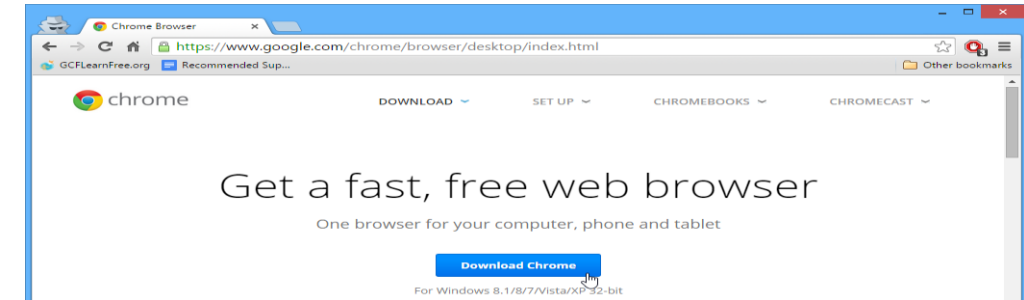
- In the web client, right-click the folder or file that you want to cut, copy, paste, rename, or delete.
- Click Cut, Copy, or Paste, Rename, or Delete as appropriate.
- If you are renaming a file or folder, type the new name for the file or folder when prompted.
- Click Save.



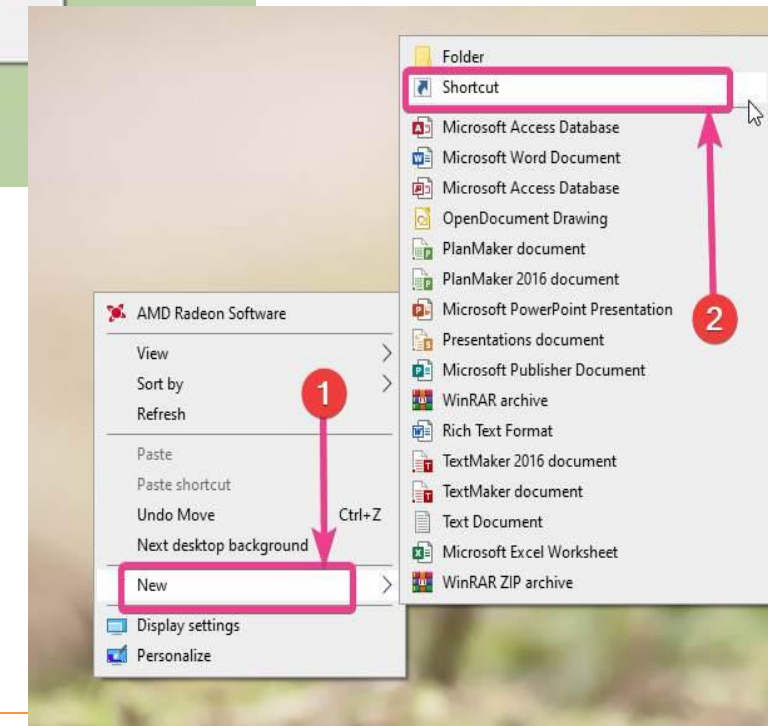
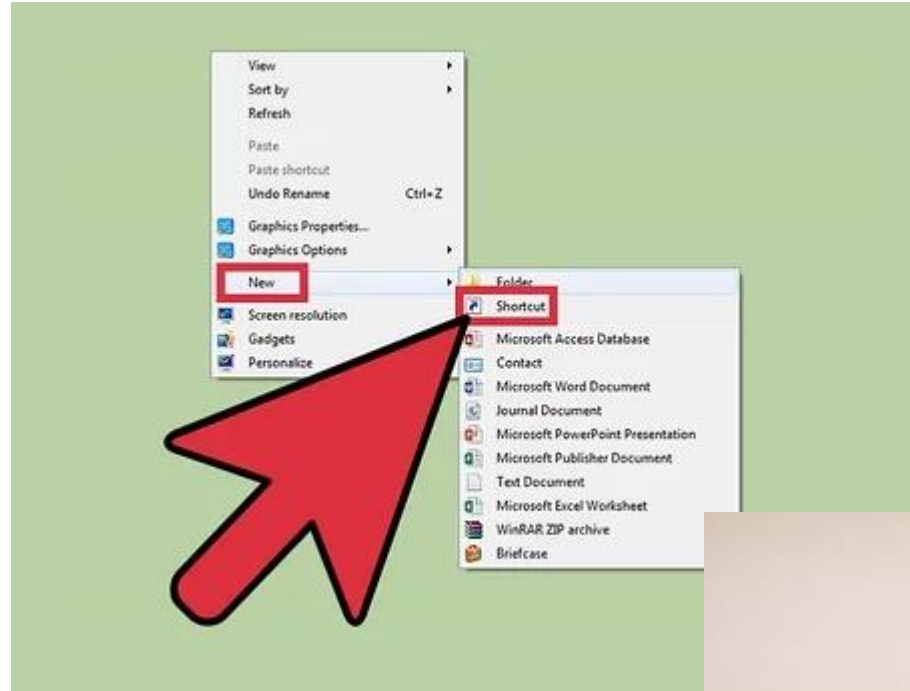
- In the desktop, click or tap the **File Explorer** button on the taskbar.
- Open an Explorer window in the location where you want to search.
- Click or tap in the Search box. A search tab appears. Disregard the tab for a simple search.
- Type a word or part of a word.
- To clear and close the search, click or tap the **Close** button (x) on the Search tab or in the Search box.



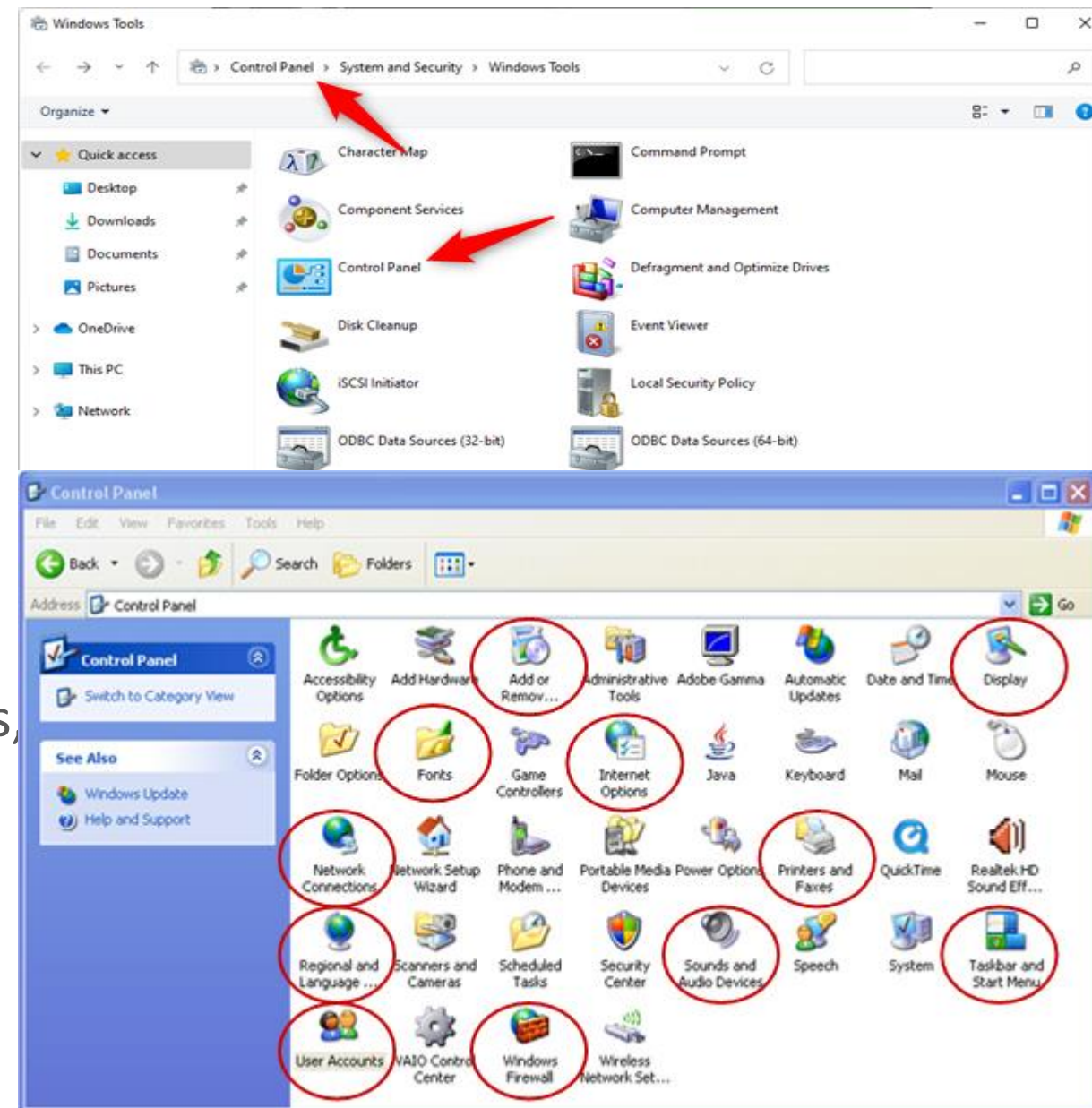
- the most common way to get new software is to **download it** from the Internet. Applications like Microsoft Office and Adobe Photoshop can now be purchased and downloaded right to your computer. You can also install free software this way. For example, if you wanted to install the Google Chrome web browser
- Locate and download an **.exe** file.
- Locate and **double-click** the .exe file. (It will usually be in your **Downloads** folder.)
- A dialog box will appear. Follow the instructions to install the software.
- The software will be installed. You can now open the application from the **Start menu** (Windows 7) or the **Start Screen** (Windows 8).



- **Desktop apps only**
- Select the Windows button to open the Start menu. ...
- Select All apps. ...
- Right-click on the app you want to create a desktop shortcut for.
- Select More.
- Select Open file location. ...
- Right-click on the app's icon.
- Select Create shortcut.
- Select Yes.



- Control Panel. The Control Panel is a **component of Microsoft Windows that provides the ability to view and change system settings.**
- It consists of a set of applets that include adding or removing hardware and software, controlling user accounts, changing accessibility options, and accessing networking settings.
- The control panel, in the context of Windows, is a **software module used to perform administrative and management operating system tasks and/or provide access to specific software features.**

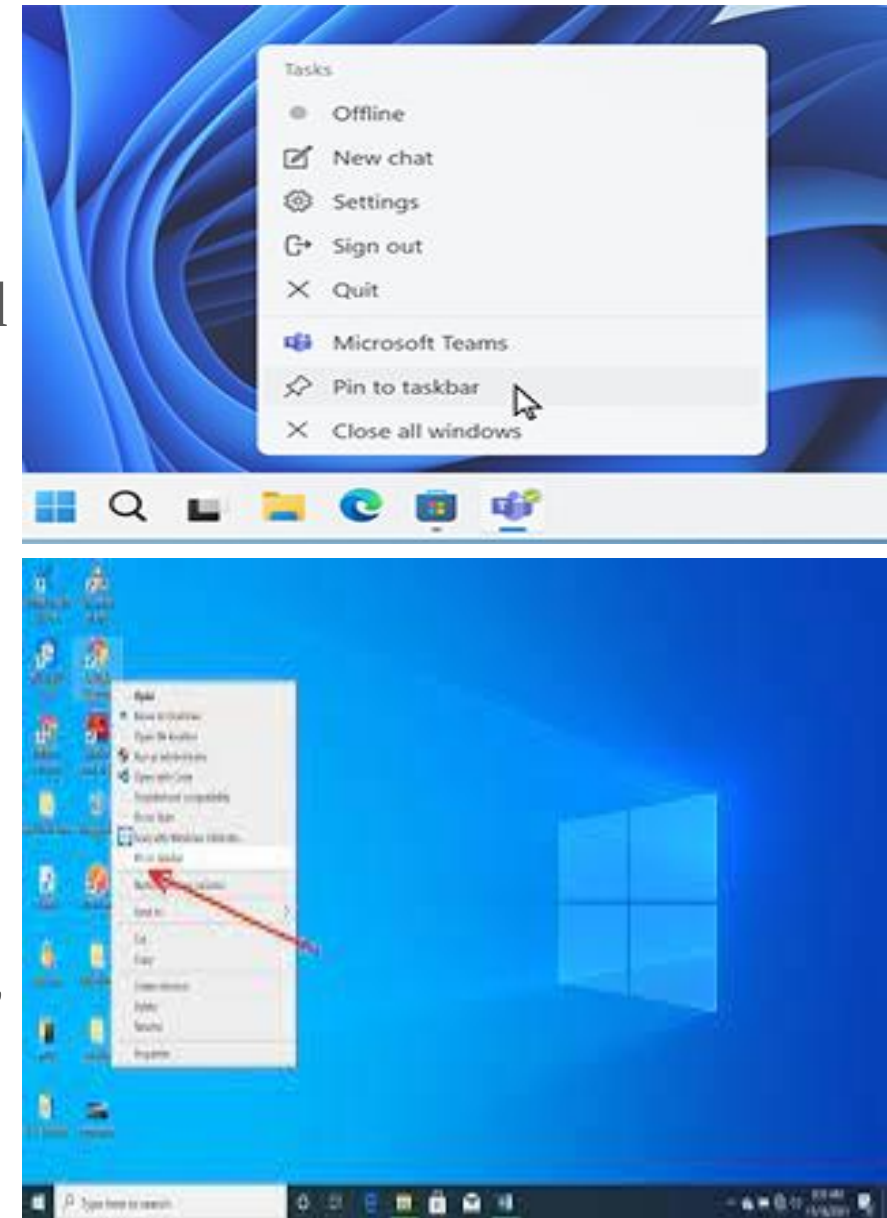


- **How to open the Control Panel in Windows Vista and 7**
- On the Windows desktop screen, click Start.
- Click Control Panel on the right side of the Start Menu.
- You may see a window similar to the following image. You may also see the expanded version of the Control Panel, with icons for all the various utilities available.



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- The taskbar on a computer screen is the narrow strip of icons, usually located at the bottom of the screen, that shows you which windows are currently open and that allows you to control functions such as the Start button and the clock. Compare toolbar.
- The Start Button--Opens the menu.
- The Quick Launch bar--contains shortcuts to commonly used applications. ...
- The main Taskbar--displays icons for all open applications and files.
- **Press and hold or right-click any empty space on the taskbar, select Taskbar settings, select Taskbar behaviors, then choose Select the far corner of the taskbar to show the desktop.** Move the mouse over, or press, the far-right edge of the taskbar to see the desktop.





Thank You!